DATA-537 NFL Prediction

By: Grant Gonnerman

Introduction

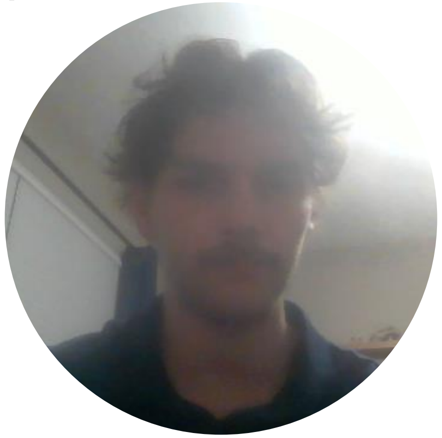
The NFL has seen massive growth in recent years.

Expanding into other countries for international games.

Setting new viewership records during holiday games.

AWS along with many other companies are providing real time statistics adding another layer to the game.

I will look at weekly team statistics with the goal of identifying the most important features to predict team wins and losses.



Data Description

I collected this data from Yahoo sports.

There are no ethical concerns with this data because it is all publicly available with no confidential information.

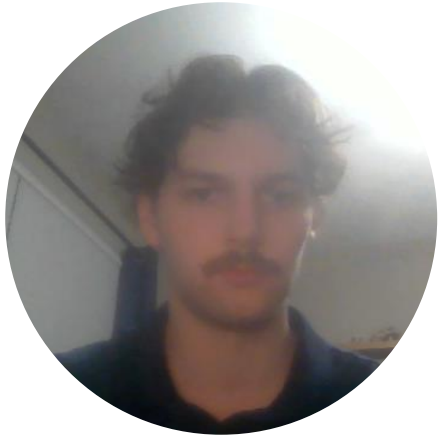
Data was saved into excel files.

Aggregated into team totals with python.

Collected from weeks 1-13 of the 2025 NFL season.

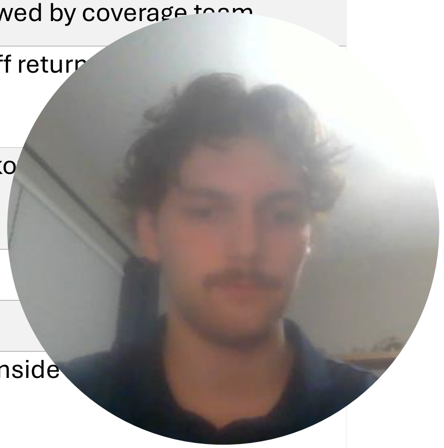
Contains defense, kicking, kickoffs, passing, punting, receiving, returns, rushing, and win/loss data.

416 rows and 79 columns in this dataset and 191 missing values.



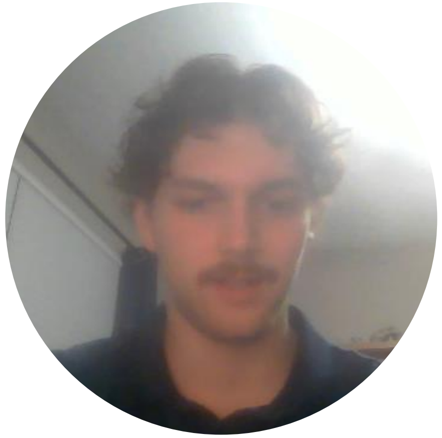
Column Name	Description
pass_completions	Number of completed passes by the quarterback
pass_attempts	Number of pass attempts thrown
passing_yards	Total yards gained through passing
passing_touchdowns	Touchdowns scored via passing plays
interceptions_thrown	Number of passes intercepted by the defense
passing_first_downs	Number of first downs gained by passing
times_sacked	Number of times the quarterback was sacked
sack_yards_lost	Total yards lost due to sacks
pass_fumbles	Number of fumbles on passing plays
pass_fumbles_lost	Fumbles on passing plays that were lost to defense
completion_percentage	Percentage of pass attempts completed
yards_per_attempt	Average yards gained per attempted pass
rush_attempts	Number of rushing attempts
rushing_yards	Total yards gained by rushing
rushing_touchdowns	Touchdowns scored through rushing plays
rushing_first_downs	First downs gained by rushing
rush_fumbles	Number of fumbles on rushing plays
rush_fumbles_lost	Fumbles on rushing plays that were lost
longest_rush	Longest rushing play recorded
yards_per_rush	Average yards gained per rush attempt
receptions	Number of passes caught
targets	Number of pass attempts thrown toward the receiver
receiving_yards	Total yards gained through receiving
receiving_touchdowns	Touchdowns scored on receptions
receiving_first_downs	First downs gained via receptions
receiving_fumbles	Fumbles on completed receptions
receiving_fumbles_lost	Receiving fumbles that were lost

yards_per_reception	**Average yards gained per catch**
fg_made_(0-19)	Field goals made from 0–19 yards
fg_attempted_(0-19)	Field goal attempts from 0–19 yards
fg_made_(20-29)	Field goals made from 20–29 yards
fg_attempted_(20-29)	Field goal attempts from 20–29 yards
fg_made_(30-39)	Field goals made from 30–39 yards
fg_attempted_(30-39)	Field goal attempts from 30–39 yards
fg_made_(40-49)	Field goals made from 40–49 yards
fg_attempted_(40-49)	Field goal attempts from 40–49 yards
fg_made_(50+)	Field goals made from 50 yards or more
fg_attempted_(50+)	Field goal attempts from 50 yards or more
fg_made_(total)	Total field goals made
fg_attempted_(total)	Total field goal attempts
extra_points_made	Extra points made after touchdowns
extra_points_attempted	Extra point attempts after touchdowns
kickoffs	Number of kickoffs made
kickoff_yards	Total yards on all kickoffs
kickoff_fair_catches	Fair catches made on kickoffs
touchbacks	Number of touchbacks on kickoffs
kick_returns_allowed	Number of kickoff returns allowed by coverage team
kick_return_touchdowns_allowed	Touchdowns allowed on kickoff retur
kick_return_yards_allowed_per_return	Average yards allowed per kicko
punts	Number of punts
punt_yards	Total yards gained on punts
punts_inside_20	Number of punts that landed inside yard line



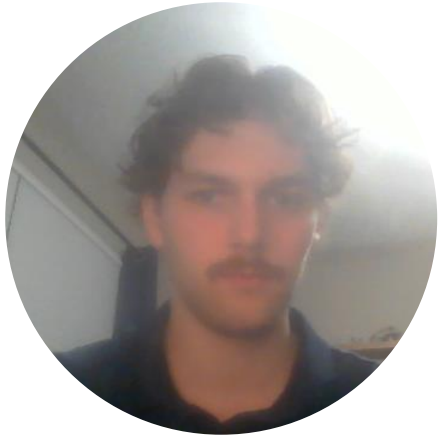
punt_fair_catches	**Number of fair catches on punts**
punt_touchbacks	Touchbacks resulting from punts
blocked_punts	Number of punts blocked by defense
punt_returns_allowed	Punt returns allowed by coverage team
punt_return_yards_allowed_per_return	Average yards allowed per punt return
kick_returns	Number of kick returns by the team
kick_return_yards	Total yards gained on kick returns
kick_return_touchdowns	Touchdowns scored on kick returns
punt_returns	Number of punt returns
average_punt_return_yards	Average yards gained per punt return
solo_tackles	Number of solo tackles
assisted_tackles	Number of assisted tackles
total_tackles	Total tackles (solo + assisted)
sacks	Number of quarterback sacks
sack_yards	Yards lost by opponent due to sacks
run_stuffs	Number of run plays stopped at or behind line of scrimmage
interceptions	Passes intercepted by defense
interception_return_yards	Yards gained on interception returns
interception_touchdowns	Touchdowns scored on interception returns
forced_fumbles	Fumbles forced on opponents
fumble_return_touchdowns	Touchdowns scored on fumble recoveries
passes_defended	Pass breakups or deflections
safeties	Safeties scored by the defense
team_name_full	Full team name (e.g., “Arizona Cardinals”)

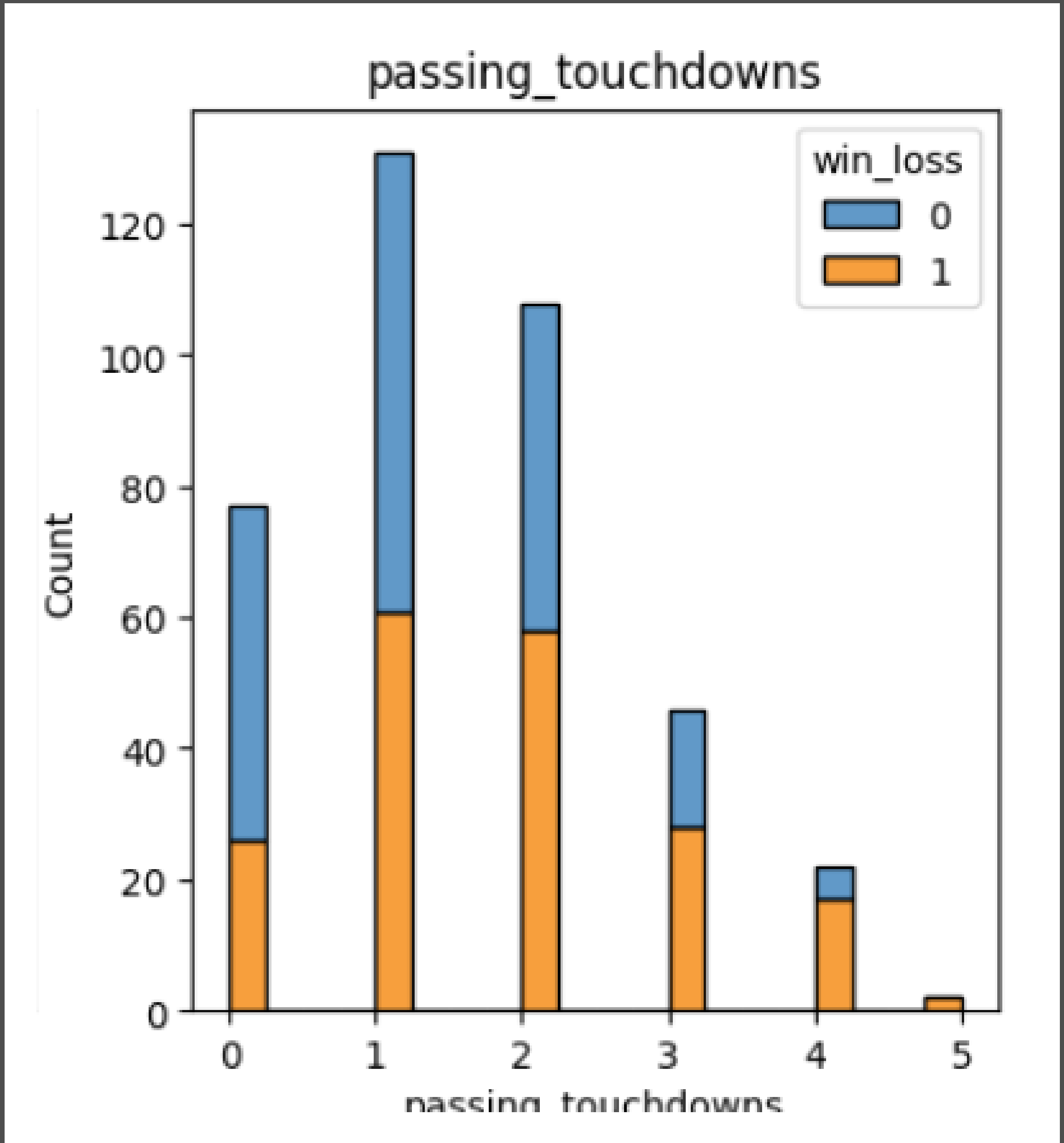
team_abbr	Team abbreviation (e.g., “ARI”)
win_loss	Indicates whether the team won or lost the game (0=loss, 1=win)
game_week	Week number of the NFL season

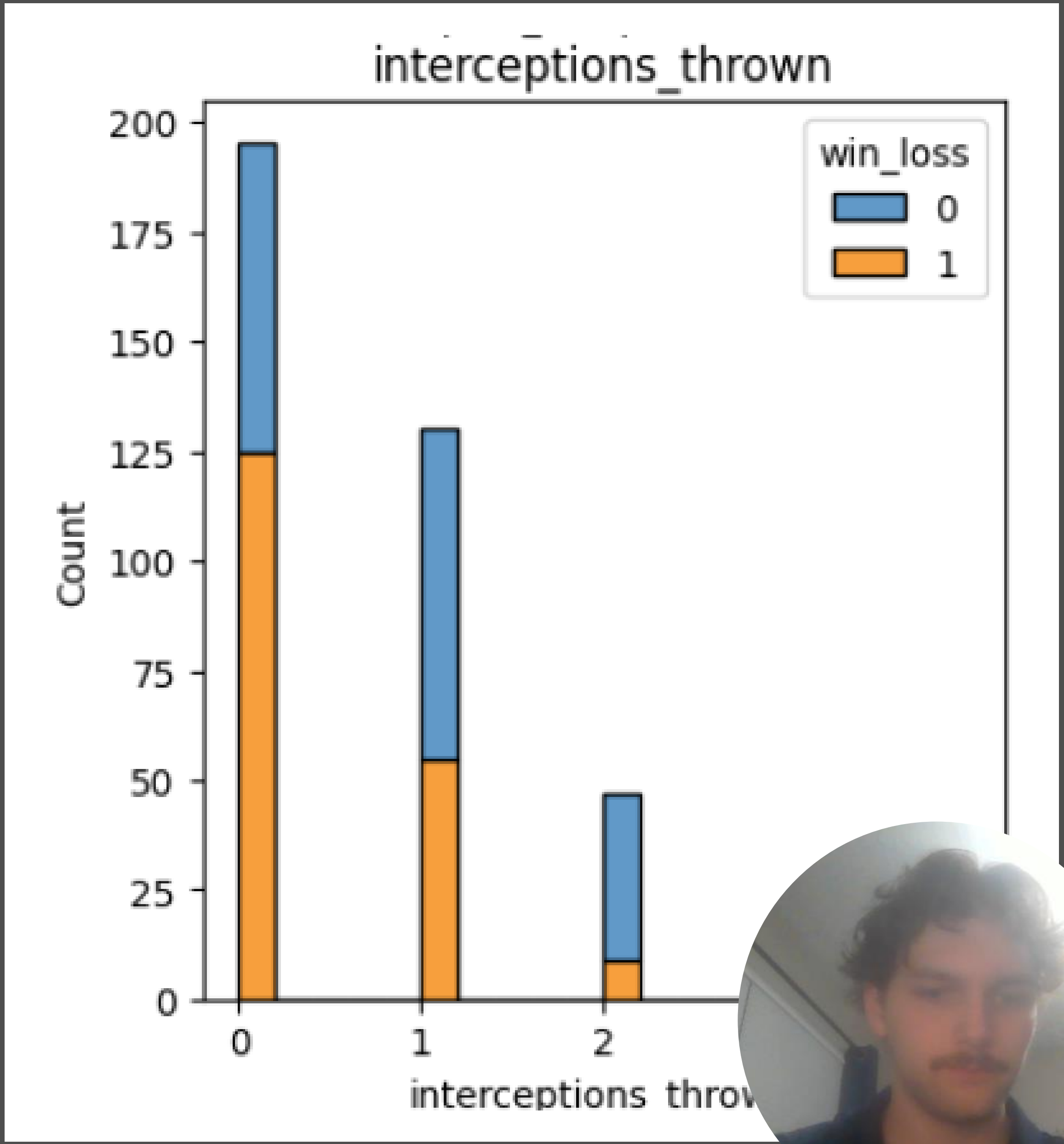


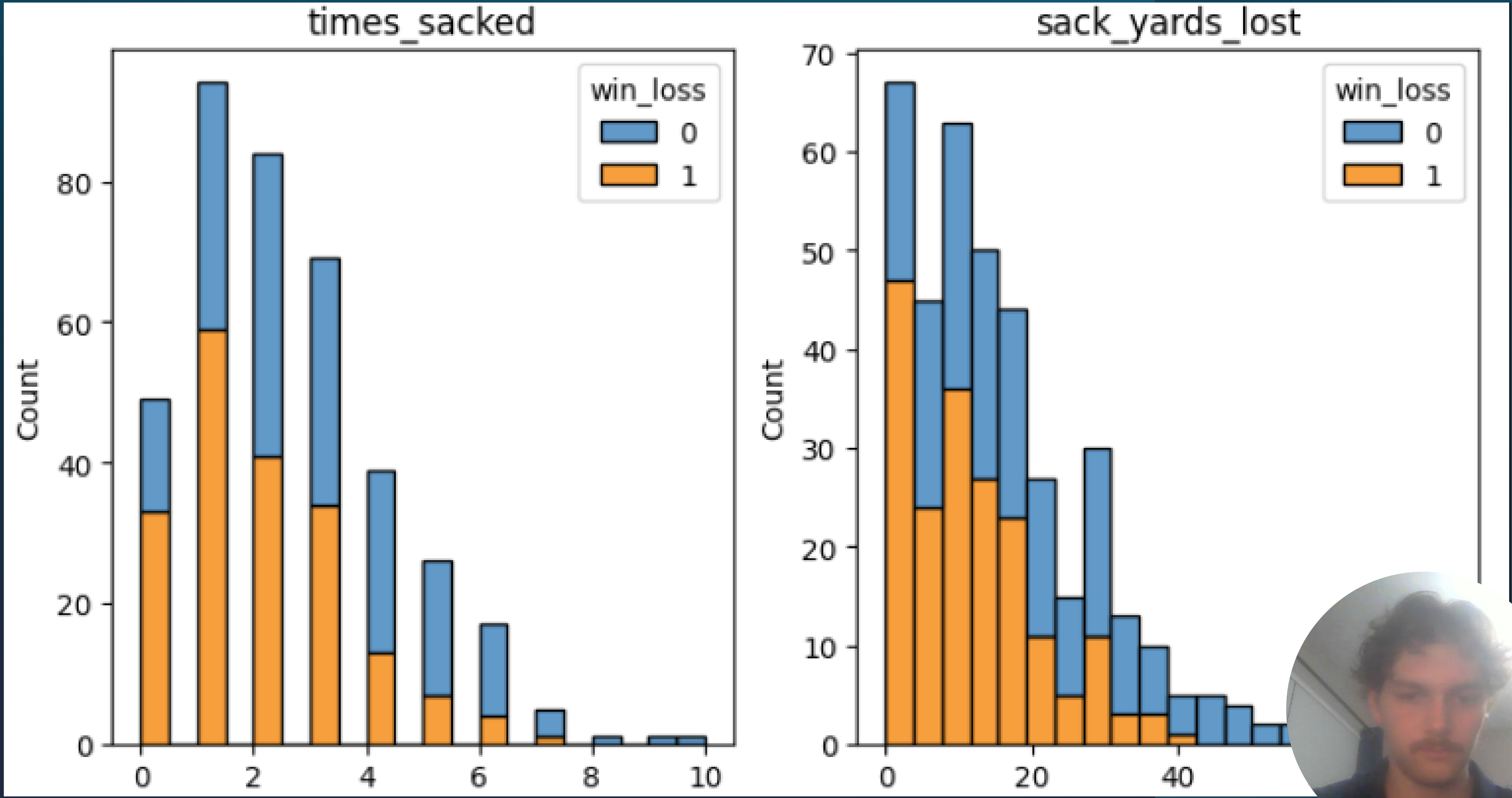
Exploratory Data Analysis

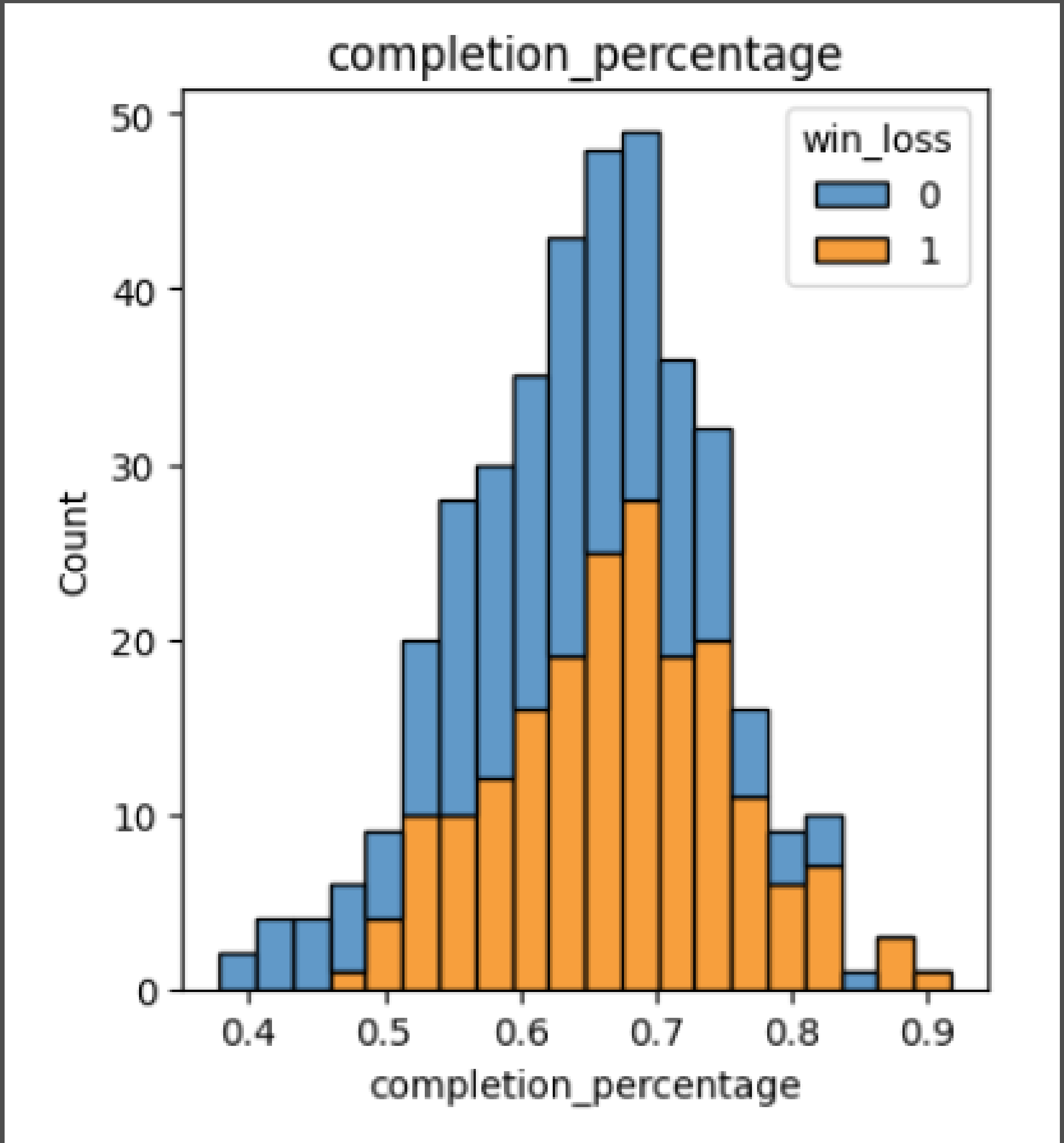
- I created a series of visualizations to explore the distributions of team statistics.
- The goal of this analysis is to identify key statistics that could be the most important factors in predicting wins and losses in the NFL.

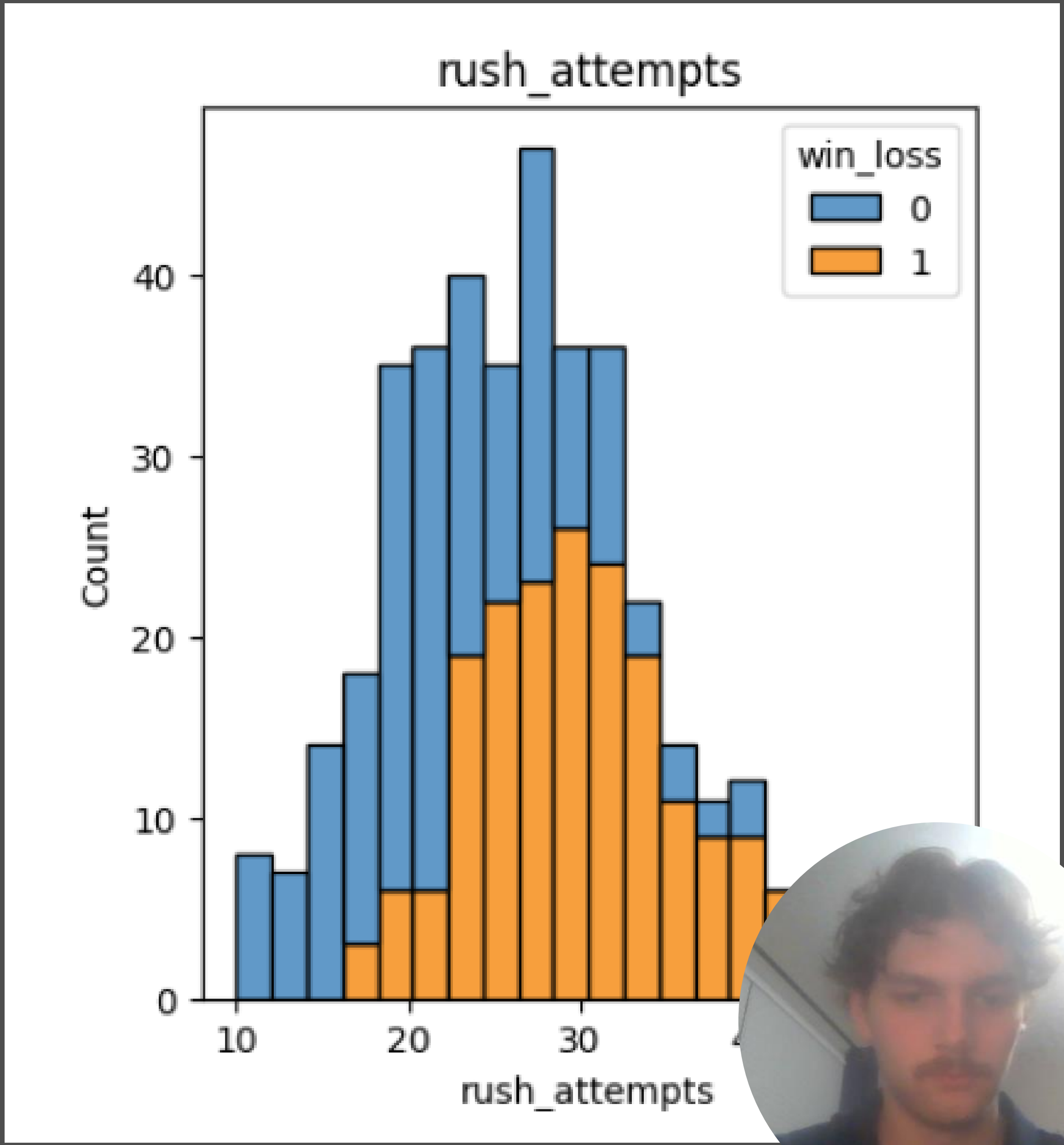


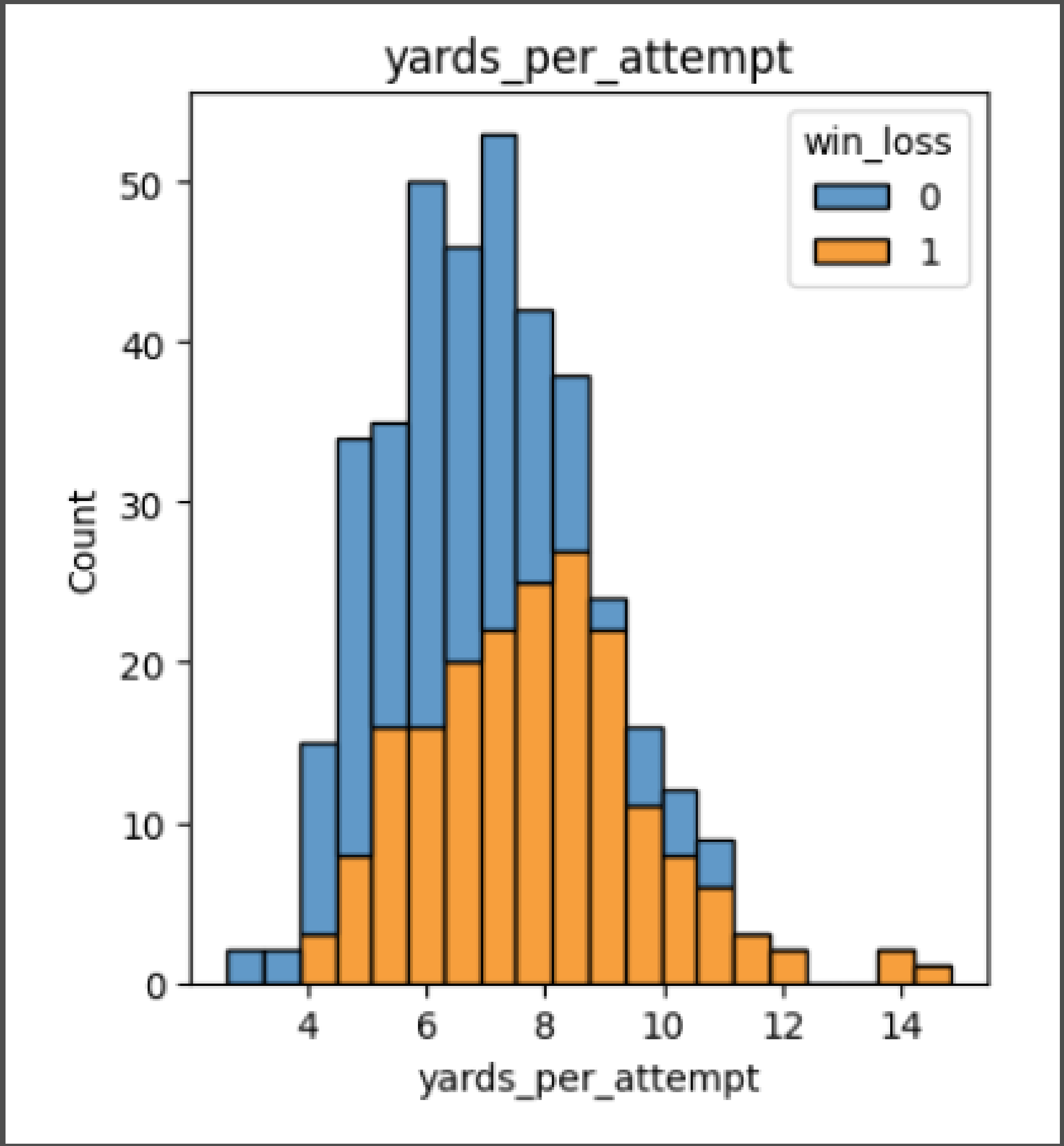

passing_touchdowns
win_loss
0
1
Count
0
20
40
60
80
100
120
0
1
2
3
4
5
passing_touchdowns

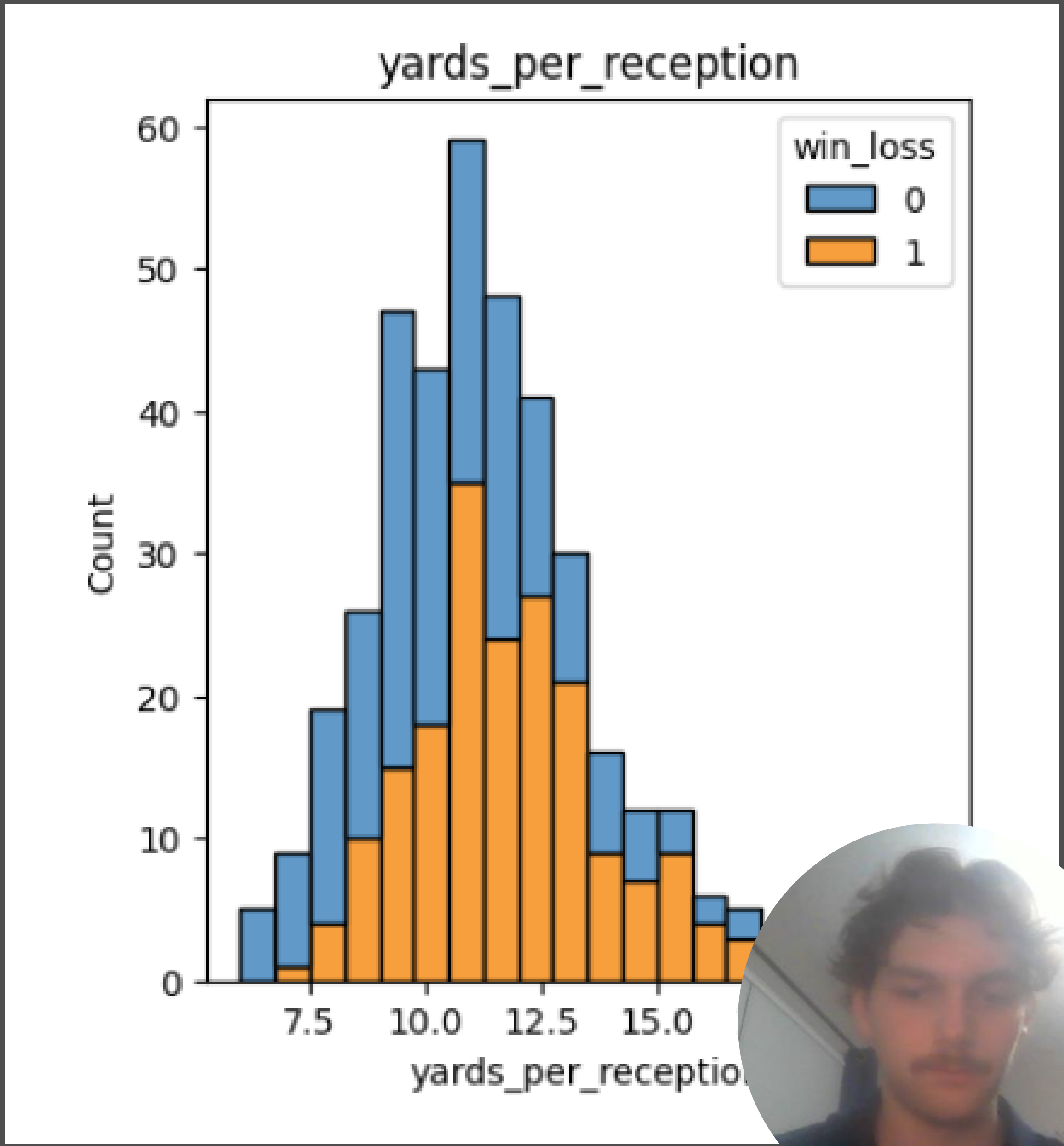

interceptions_thrown
win_loss
0
1
Count
0
25
50
75
100
125
150
175
200
0
1
2

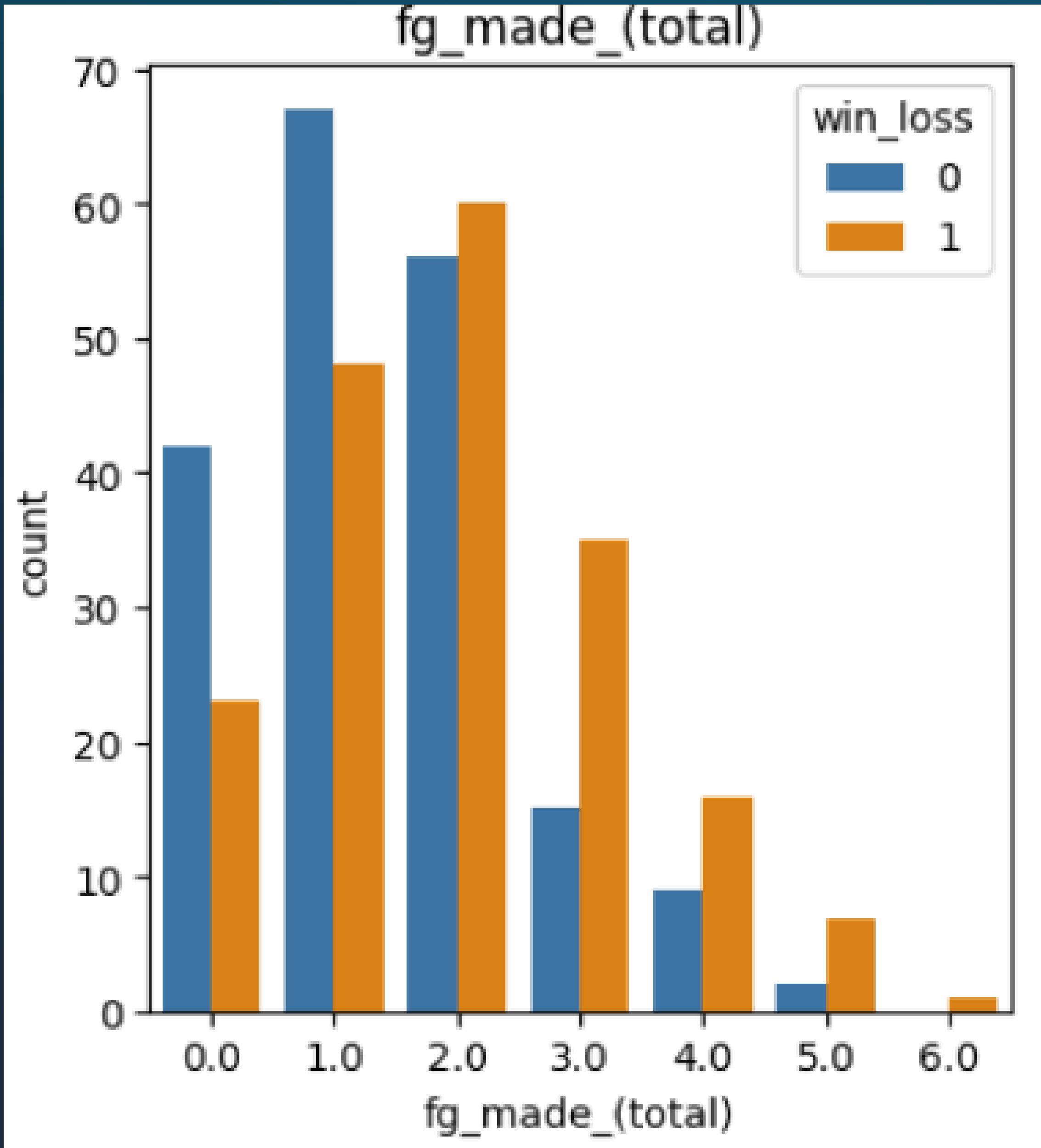

times_sacked
sack_yards_lost
Count
Count
win_loss
0
1
win_loss
0
1
0
20
40
60
80
0
10
20
30
40
50
60
70
0
2
4
6
8
10
0
20
40

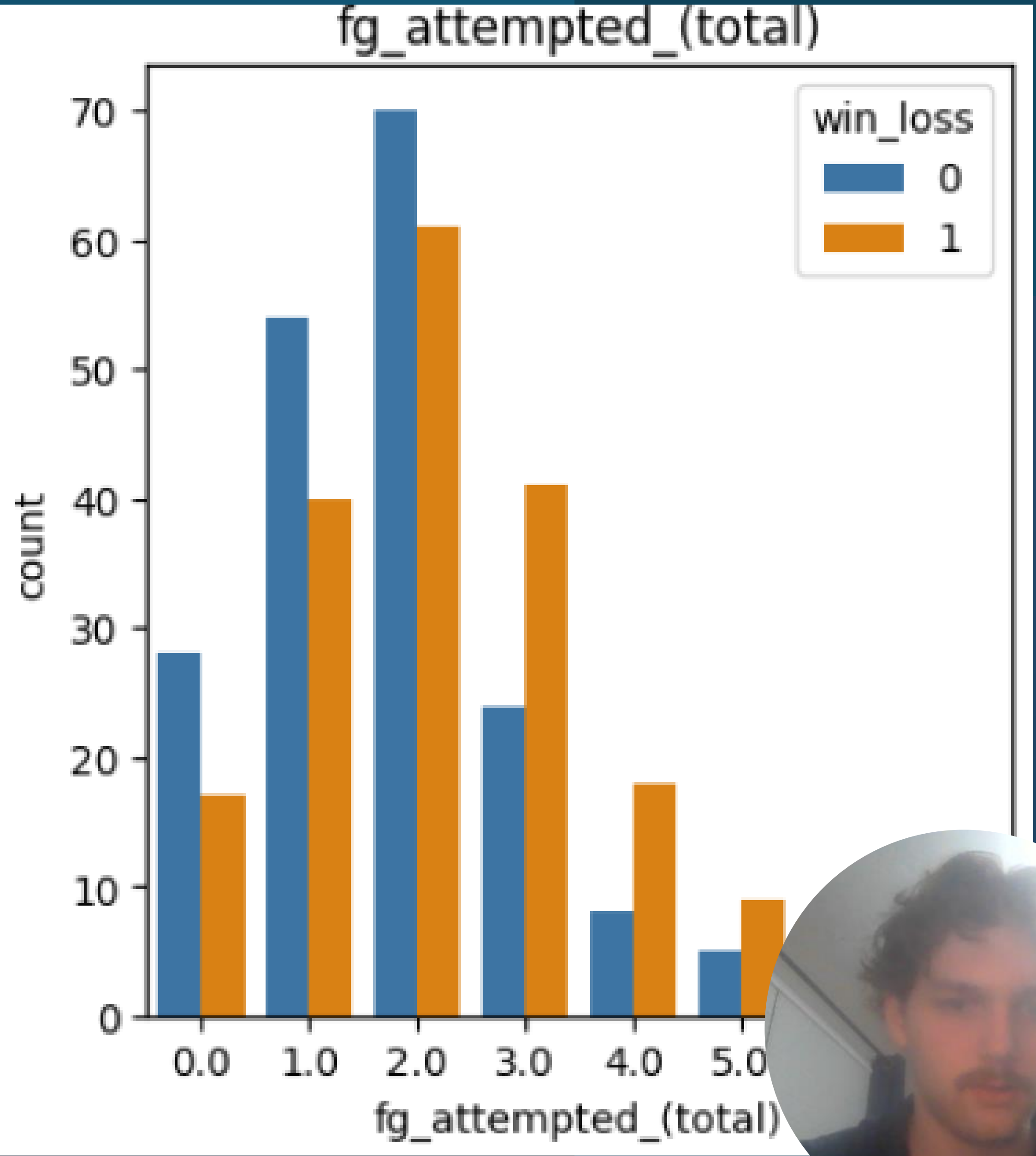

completion_percentage
win_loss
0
1
Count
0
10
20
30
40
50
0.4
0.5
0.6
0.7
0.8
0.9
completion_percentage

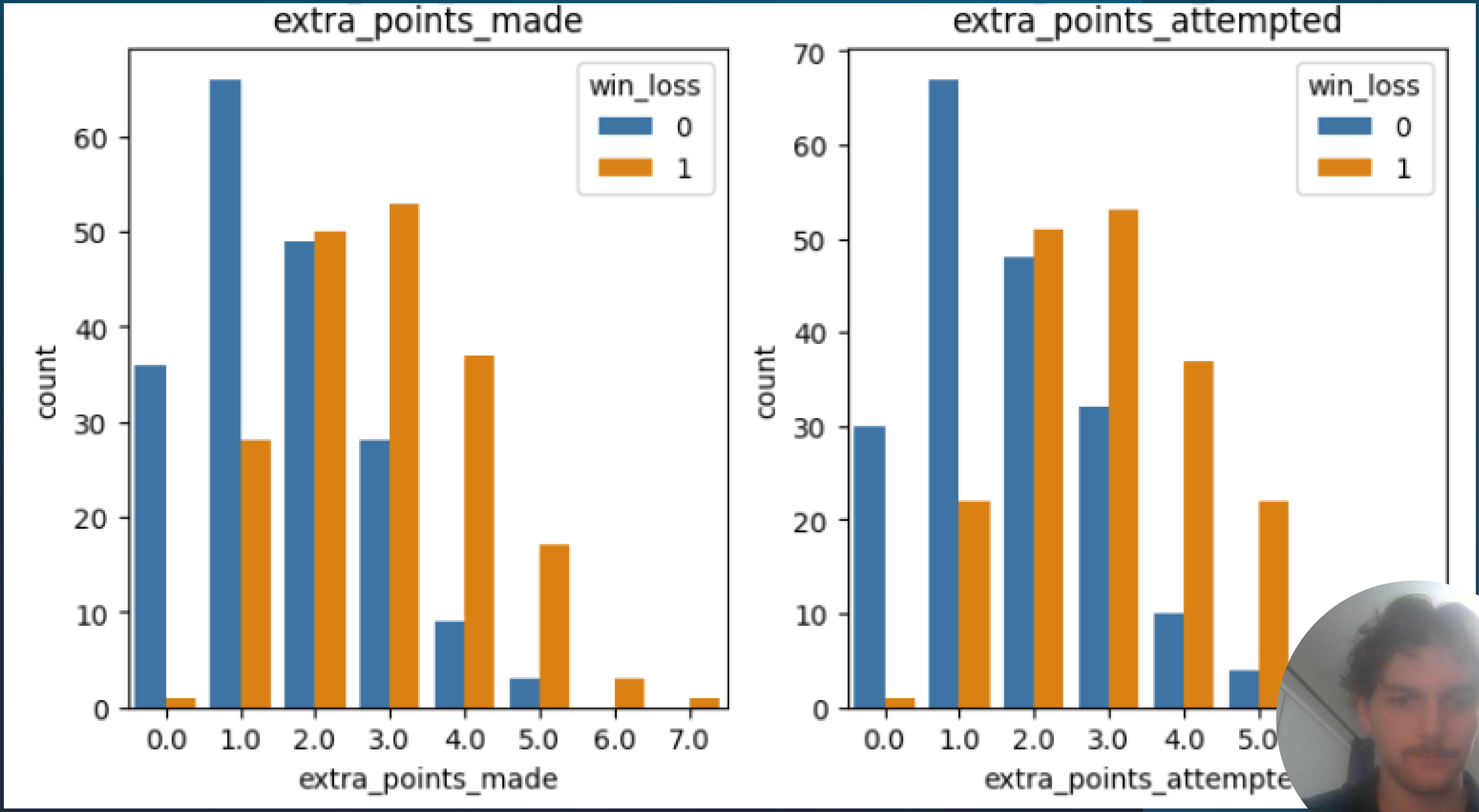

rush_attempts
win_loss
0
1
Count
0
10
20
30
40
10
20
30
rush_attempts

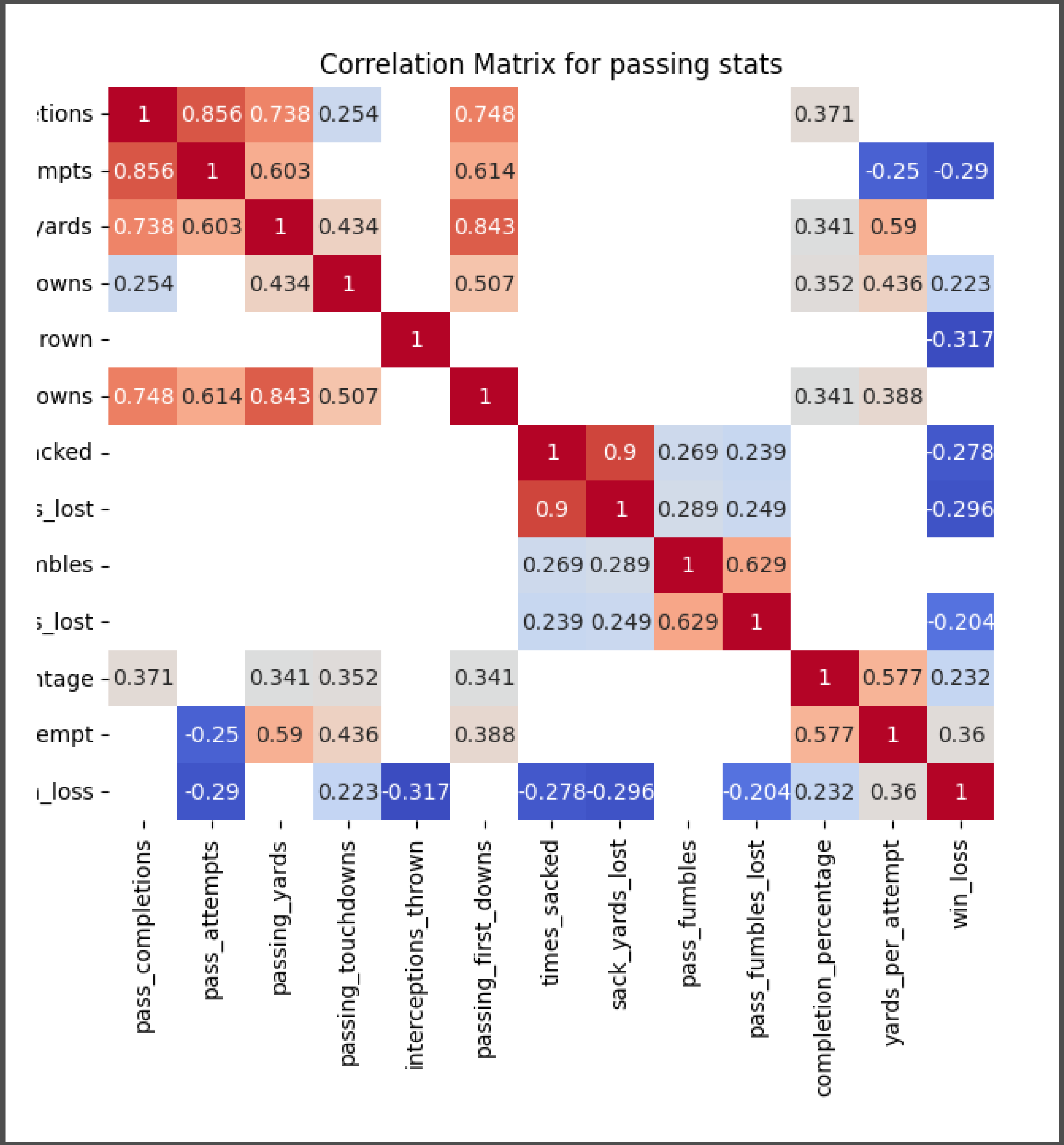

yards_per_attempt
win_loss
0
1
Count
yards_per_attempt

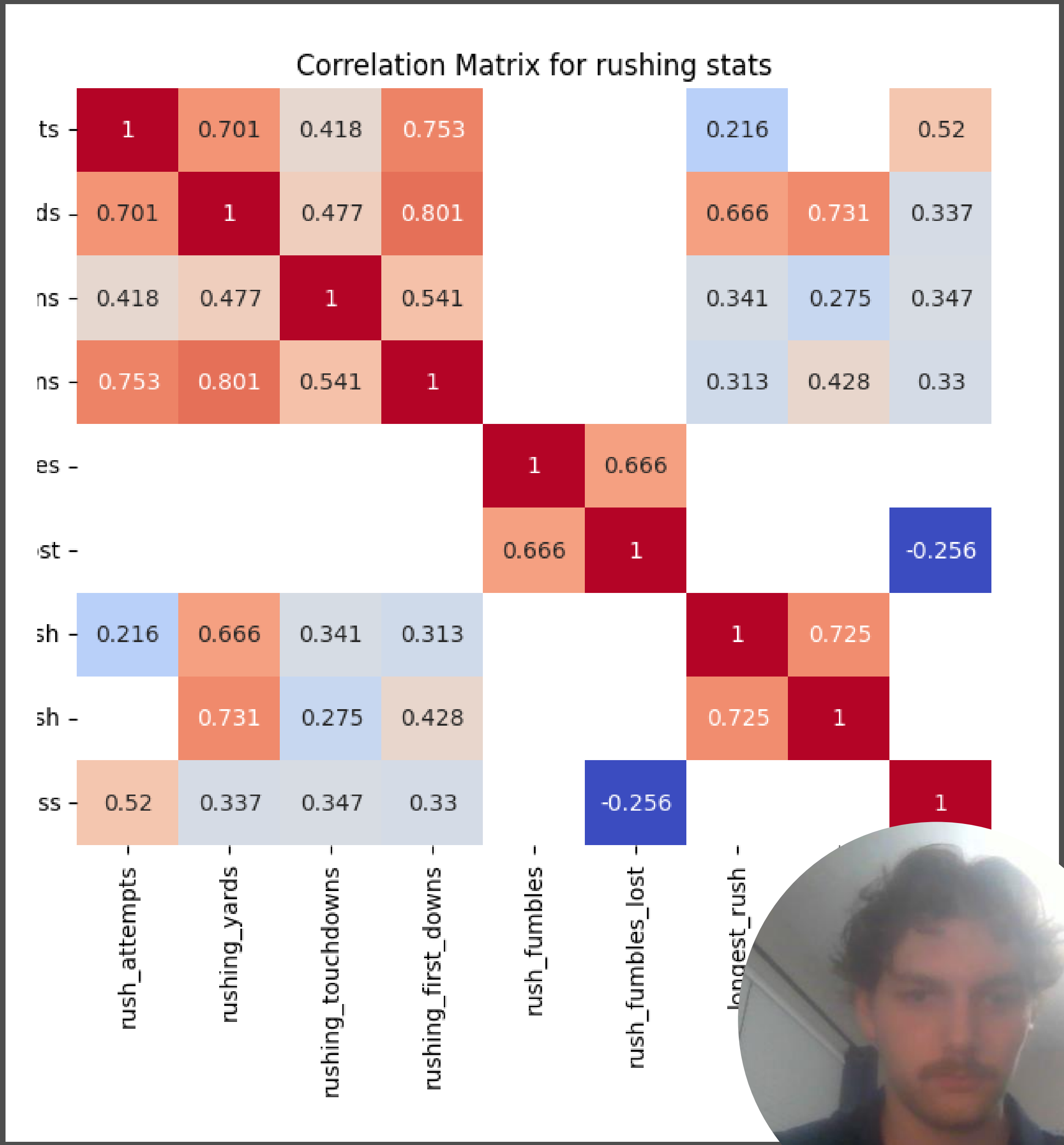

yards_per_reception
win_loss
0
1
Count

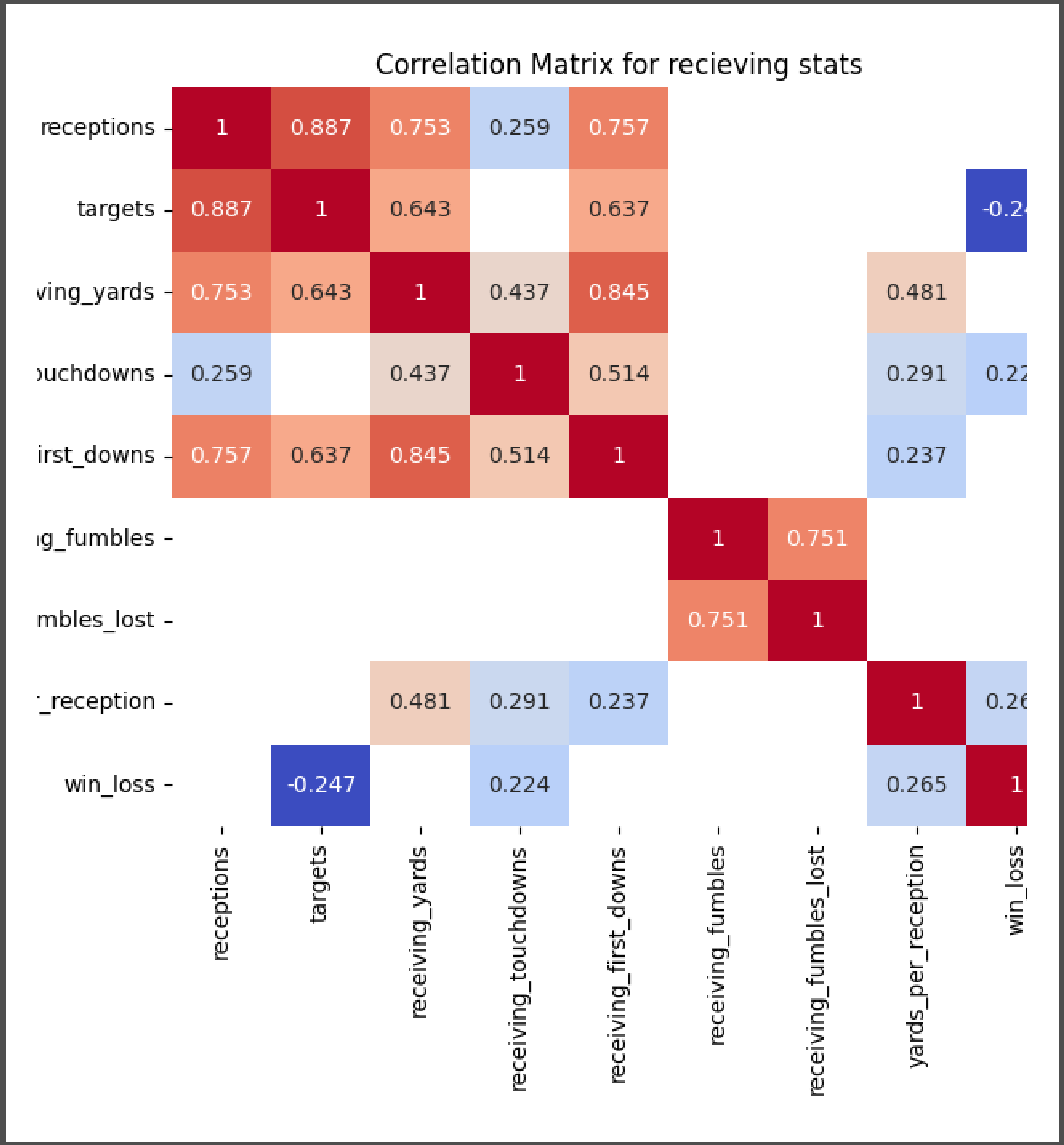

fg_made_(total)
count
70
60
50
40
30
20
10
0
0.0
1.0
2.0
3.0
4.0
5.0
6.0
fg_made_(total)
win_loss
0
1

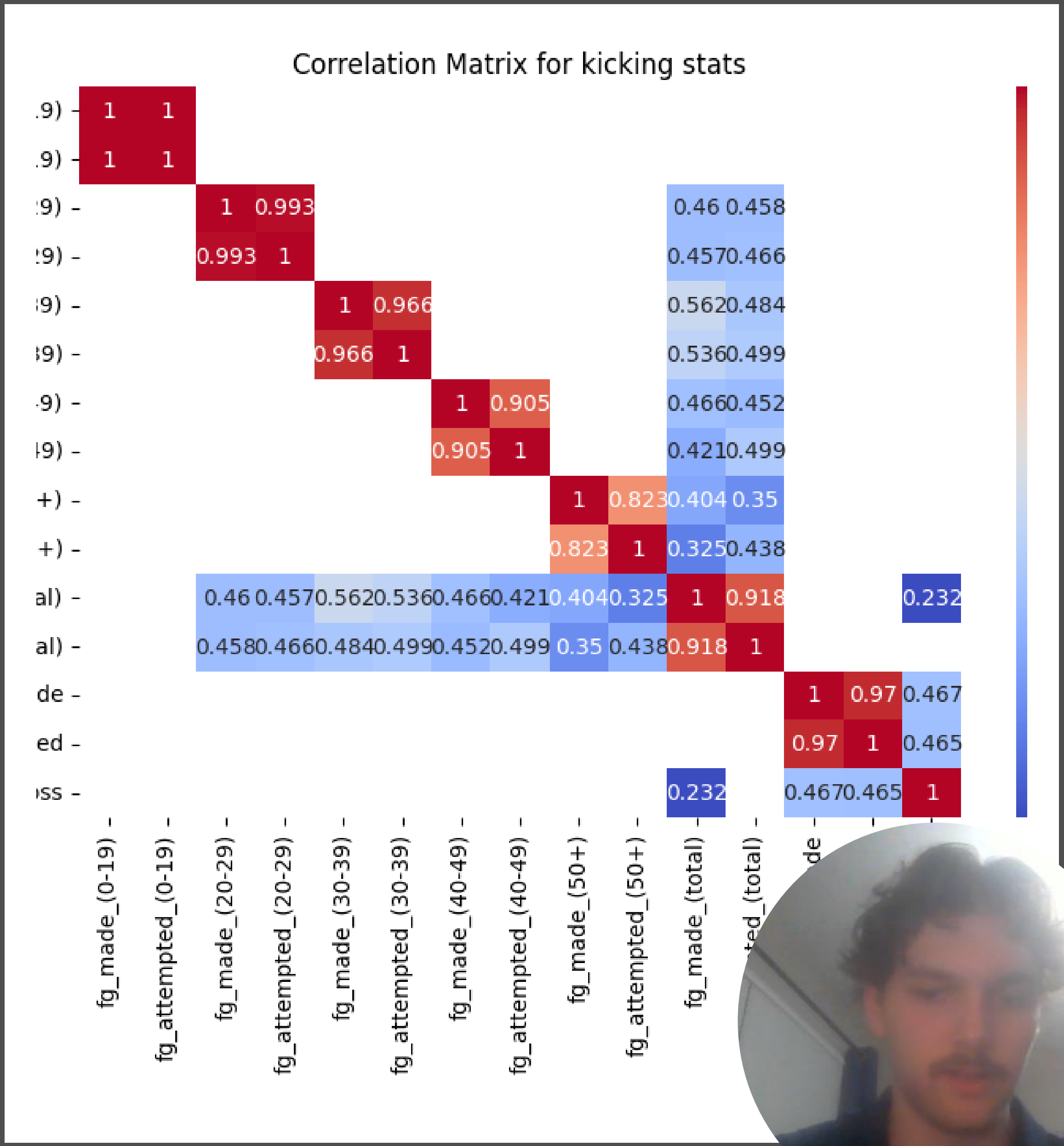

fg_attempted_(total)
count
70
60
50
40
30
20
10
0
0.0
1.0
2.0
3.0
4.0
5.0
fg_attempted_(total)
win_loss
0
1

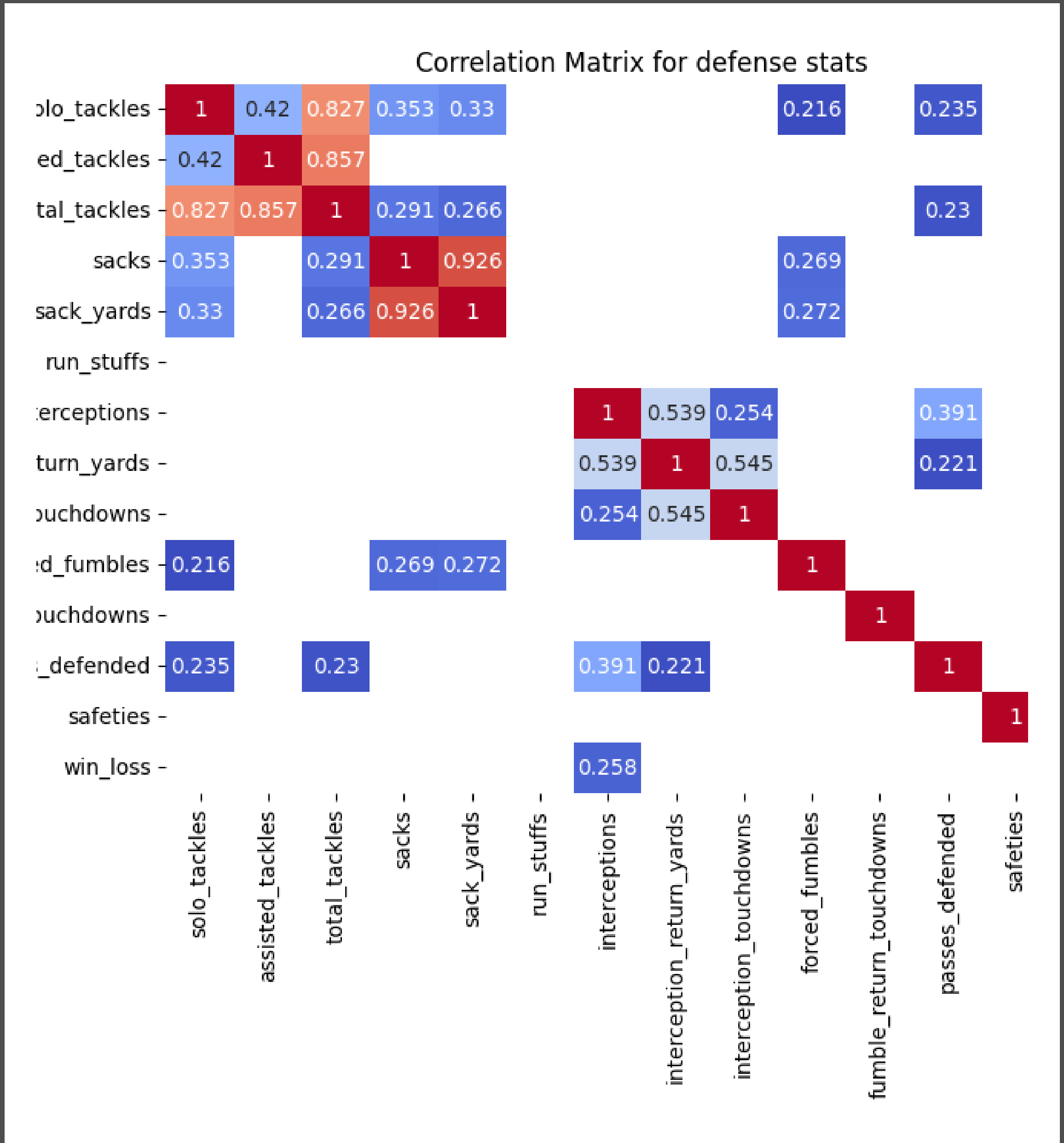

extra_points_made
win_loss
0
1
count
60
50
40
30
20
10
0
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
extra_points_made
extra_points_attempted
win_loss
0
1
count
70
60
50
40
30
20
10
0
0.0
1.0
2.0
3.0
4.0
5.0
extra_points_attempte

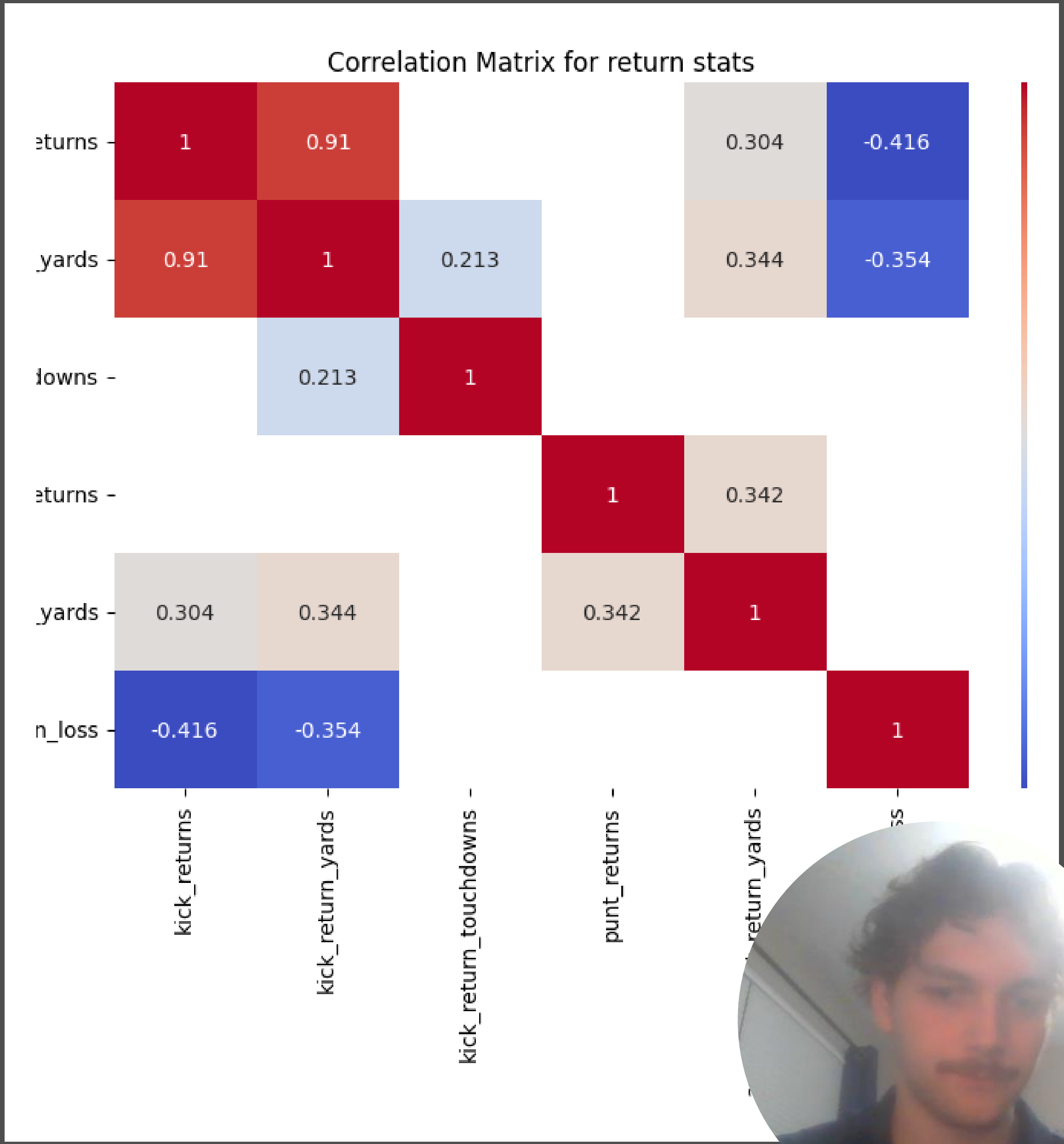

Correlation Matrix for passing stats
pass_completions
pass_attempts
passing_yards
passing_touchdowns
interception_thrown
passing_first_downs
times_sacked
sack_yards_lost
pass_fumbles
pass_fumbles_lost
completion_percentage
yards_per_attempt
win_loss


Correlation Matrix for rushing stats
rush_attempts
rushing_yards
rushing_touchdowns
rushing_first_downs
rush_fumbles
rush_fumbles_lost


Correlation Matrix for recieving stats
receptions
targets
receiving_yards
receiving_touchdowns
receiving_first_downs
receiving_fumbles
receiving_fumbles_lost
yards_per_reception
win_loss

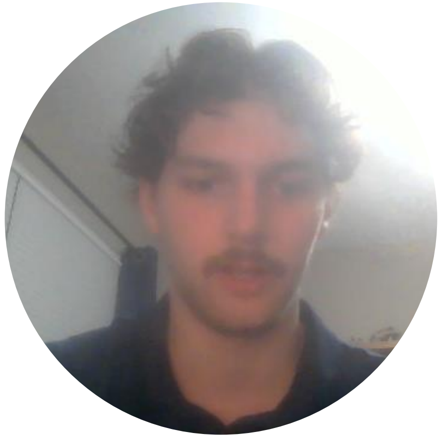

Correlation Matrix for kicking stats
fg_made_(0-19)
fg_attempted_(0-19)
fg_made_(20-29)
fg_attempted_(20-29)
fg_made_(30-39)
fg_attempted_(30-39)
fg_made_(40-49)
fg_attempted_(40-49)
fg_made_(50+)
fg_attempted_(50+)
fg_made_(total)


Correlation Matrix for defense stats
olo_tackles
ed_tackles
tal_tackles
sacks
sack_yards
run_stuffs
erceptions
turn_yards
ouchdowns
ed_fumbles
ouchdowns
_defended
safeties
win_loss
solo_tackles
assisted_tackles
total_tackles
sacks
sack_yards
run_stuffs
interceptions
interception_return_yards
interception_touchdowns
forced_fumbles
fumble_return_touchdowns
passes_defended
safeties


Correlation Matrix for return stats
eturns
yards
downs
eturns
yards
n_loss
kick_returns
kick_return_yards
kick_return_touchdowns
punt_returns
return_yards

Modeling

- There were missing values in this dataset.
  - Filled with 0.
- Built a logistic regression model on each of the statistics groups.
- After running each model, I would add significant columns to a list to use in my final model.



Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	377
Method:	MLE	Df Model:	8
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1746
Time:	17:35:41	Log-Likelihood:	-220.82
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	9.240e-17

Logit Regression Results

	coef	std err	z	P>|z|	[0.025	0.975]
const	3.1901	2.065	1.545	0.122	-0.858	7.238
receptions	-0.1085	0.119	-0.908	0.364	-0.343	0.126
targets	-0.1580	0.036	-4.383	0.000	-0.229	-0.087
receiving_yards	0.0118	0.010	1.167	0.243	-0.008	0.032
receiving_touchdowns	0.1344	0.125	1.078	0.281	-0.110	0.379
receiving_first_downs	0.2379	0.074	3.218	0.001	0.093	0.383
receiving_fumbles	-0.2795	0.269	-1.038	0.299	-0.807	0.248
receiving_fumbles_lost	-0.4092	0.411	-0.997	0.319	-1.214	0.395
yards_per_reception	-0.1484	0.180	-0.822	0.411	-0.502	0.205

Logit Regression Results

Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	377
Method:	MLE	Df Model:	8
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.3222
Time:	17:34:57	Log-Likelihood:	-181.36
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	4.077e-33

	coef	std err	z	P>|z|	[0.025	0.975]
const	-5.8326	1.842	-3.167	0.002	-9.442	-2.223
rush_attempts	0.2767	0.075	3.688	0.000	0.130	0.424
rushing_yards	-0.0115	0.016	-0.727	0.468	-0.043	0.020
rushing_touchdowns	0.6327	0.175	3.624	0.000	0.291	0.975
rushing_first_downs	-0.1782	0.093	-1.916	0.055	-0.360	0.004
rush_fumbles	-0.0680	0.204	-0.333	0.739	-0.468	0.332
rush_fumbles_lost	-1.0842	0.322	-3.365	0.001	-1.716	-0.453
longest_rush	0.0288	0.015	1.867	0.062	-0.001	0.059
yards_per_rush	0.0454	0.394	0.115	0.908	-0.727	0.817

Logit Regression Results

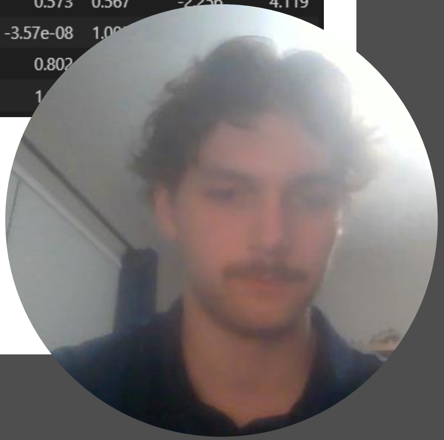
Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	373
Method:	MLE	Df Model:	12
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2826
Time:	17:34:50	Log-Likelihood:	-191.94
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	3.221e-26

	coef	std err	z	P>|z|	[0.025	0.975]
const	-0.9969	4.219	-0.236	0.813	-9.266	7.273
pass_completions	-0.1830	0.246	-0.743	0.457	-0.666	0.300
pass_attempts	-0.0099	0.131	-0.075	0.940	-0.266	0.247
passing_yards	-0.0045	0.012	-0.372	0.710	-0.028	0.019
passing_touchdowns	-0.0305	0.139	-0.219	0.827	-0.303	0.242
interceptions_thrown	-0.7151	0.169	-4.228	0.000	-1.047	-0.384
passing_first_downs	0.2911	0.081	3.597	0.000	0.132	0.450
times_sacked	-0.0848	0.166	-0.512	0.609	-0.409	0.240
sack_yards_lost	-0.0326	0.024	-1.370	0.171	-0.079	0.014
pass_fumbles	0.3221	0.242	1.330	0.183	-0.153	0.797
pass_fumbles_lost	-1.2281	0.406	-3.027	0.002	-2.023	-0.433
completion_percentage	3.8204	7.913	0.483	0.629	-11.689	19.330
yards_per_attempt	0.2538	0.361	0.704	0.482	-0.453	0.961

Logit Regression Results

Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	374
Method:	MLE	Df Model:	11
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2699
Time:	17:35:47	Log-Likelihood:	-195.33
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	2.036e-25

	coef	std err	z	P>|z|	[0.025	0.975]
const	-3.3497	0.412	-8.126	0.000	-4.158	-2.542
fg_made_(0-19)	0.0003	nan	nan	nan	nan	nan
fg_attempted_(0-19)	0.0003	nan	nan	nan	nan	nan
fg_made_(20-29)	0.6892	0	inf	0.000	0.689	0.689
fg_attempted_(20-29)	-0.2803	1.08e+07	-2.59e-08	1.000	-2.12e+07	2.12e+07
fg_made_(30-39)	0.5073	1.873	0.271	0.787	-3.164	4.178
fg_attempted_(30-39)	-0.2813	1.08e+07	-2.6e-08	1.000	-2.12e+07	2.12e+07
fg_made_(40-49)	-0.3309	1.529	-0.216	0.829	-3.327	2.665
fg_attempted_(40-49)	0.1389	1.08e+07	1.28e-08	1.000	-2.12e+07	2.12e+07
fg_made_(50+)	0.0658	1.583	0.042	0.967	-3.037	3.169
fg_attempted_(50+)	0.0366	1.08e+07	3.38e-09	1.000	-2.12e+07	2.12e+07
fg_made_(total)	0.9316	1.626	0.573	0.567	-2.256	4.119
fg_attempted_(total)	-0.3858	1.08e+07	-3.57e-08	[illegible]	[illegible]	[illegible]
extra_points_made	0.2957	0.369	0.802	[illegible]	[illegible]	[illegible]
extra_points_attempted	0.7258	0.372	[illegible]	[illegible]	[illegible]	[illegible]



Logit Regression Results

Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	382
Method:	MLE	Df Model:	3
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2323
Time:	17:35:56	Log-Likelihood:	-205.40
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	9.165e-27

	coef	std err	z	P>|z|	[0.025	0.975]
const	-3.8232	0.481	-7.951	0.000	-4.766	-2.881
kickoffs	-0.5393	0.346	-1.560	0.119	-1.217	0.138
kickoff_yards	0.0214	0.006	3.722	0.000	0.010	0.033
touchbacks	0.0653	0.129	0.506	0.613	-0.187	0.318

Logit Regression Results

Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	380
Method:	MLE	Df Model:	5
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1374
Time:	17:36:12	Log-Likelihood:	-230.78
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	1.878e-14

	coef	std err	z	P>|z|	[0.025	0.975]
const	2.3508	0.338	6.961	0.000	1.689	3.013
kick_returns	-0.7337	0.181	-4.062	0.000	-1.088	-0.380
kick_return_yards	0.0054	0.006	0.831	0.406	-0.007	0.018
kick_return_touchdowns	1.1431	1.033	1.106	0.269	-0.882	3.168
punt_returns	-0.1121	0.098	-1.144	0.252	-0.304	0.080
average_punt_return_yards	0.0020	0.003	0.727	0.467	-0.003	0.007

Logit Regression Results

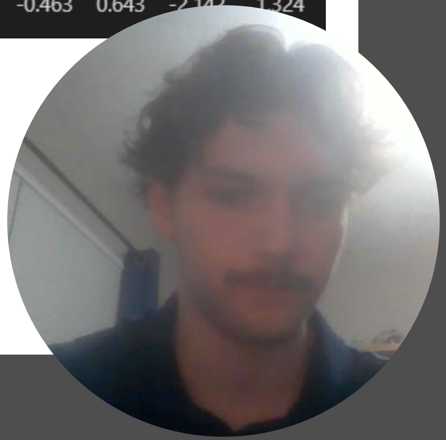
Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	373
Method:	MLE	Df Model:	12
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1538
Time:	17:36:19	Log-Likelihood:	-226.41
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	1.517e-12

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.0072	0.410	2.456	0.014	0.204	1.811
solo_tackles	-0.0280	nan	nan	nan	nan	nan
assisted_tackles	-0.0090	nan	nan	nan	nan	nan
total_tackles	-0.0369	nan	nan	nan	nan	nan
sacks	-0.2126	0.233	-0.913	0.361	-0.669	0.244
sack_yards	0.0756	0.032	2.398	0.016	0.014	0.137
interceptions	0.7278	0.253	2.879	0.004	0.232	1.223
interception_return_yards	0.0277	0.016	1.704	0.088	-0.004	0.060
interception_touchdowns	-1.9529	0.967	-2.020	0.043	-3.848	-0.058
forced_fumbles	0.3940	0.208	1.894	0.058	-0.014	0.802
fumble_return_touchdowns	1.0549	1.202	0.877	0.380	-1.302	3.412
passes_defended	0.2012	0.073	2.758	0.006	0.058	0.344
safeties	0.2648	1.366	0.194	0.846	-2.412	2.942

Logit Regression Results

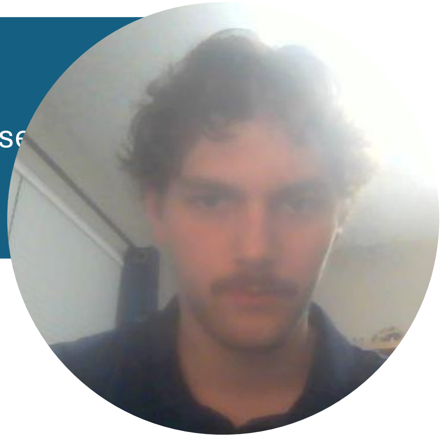
Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	380
Method:	MLE	Df Model:	5
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.01309
Time:	17:36:04	Log-Likelihood:	-264.05
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	0.2202

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.4065	0.242	1.680	0.093	-0.068	0.881
punts	-0.4747	0.289	-1.643	0.100	-1.041	0.092
punt_yards	0.0062	0.006	1.085	0.278	-0.005	0.017
punts_inside_20	0.1542	0.112	1.374	0.169	-0.066	0.374
punt_touchbacks	0.1120	0.218	0.515	0.607	-0.315	0.539
blocked_punts	-0.4093	0.884	-0.463	0.643	-2.142	1.324



Feature Selection Summary

interceptions_thrown	passing_first_downs	pass_fumbles_lost	rush_attempts	rushing_touchdowns
rush_fumbles_lost	targets	receiving_first_downs	fg_made_(20-29)	kickoff_yards
kick_returns	sack_yards	interceptions	interception_touchdowns	passe



Logit Regression Results			
Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	370
Method:	MLE	Df Model:	15
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5364
Time:	17:43:43	Log-Likelihood:	-124.03
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	2.750e-52

	coef	std err	z	P>|z|	[0.025	0.975]
const	-4.0886	1.434	-2.851	0.004	-6.899	-1.278
interceptions_thrown	-0.5415	0.224	-2.421	0.015	-0.980	-0.103
passing_first_downs	-0.5092	0.545	-0.935	0.350	-1.577	0.558
pass_fumbles_lost	-1.0107	0.620	-1.630	0.103	-2.226	0.205
rush_attempts	0.1375	0.032	4.292	0.000	0.075	0.200
rushing_touchdowns	0.2939	0.193	1.526	0.127	-0.083	0.671
rush_fumbles_lost	-0.2232	0.431	-0.518	0.605	-1.068	0.622
targets	-0.1101	0.036	-3.049	0.002	-0.181	-0.039
receiving_first_downs	0.8015	0.549	1.460	0.144	-0.275	1.878
fg_made_(20-29)	-0.0679	0.278	-0.245	0.807	-0.612	0.476
kickoff_yards	0.0075	0.002	3.545	0.000	0.003	0.012
kick_returns	-0.6198	0.115	-5.376	0.000	-0.846	-0.394
sack_yards	0.0479	0.018	2.713	0.007	0.013	0.082
interceptions	0.9244	0.316	2.924	0.003	0.305	1.544
interception_touchdowns	-1.3758	1.251	-1.099	0.272	-3.828	1.077
passes_defended	0.0952	0.094	1.011	0.312	-0.089	0.280

Logit Regression Results			
Dep. Variable:	win_loss	No. Observations:	386
Model:	Logit	Df Residuals:	376
Method:	MLE	Df Model:	9
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5232
Time:	17:52:17	Log-Likelihood:	-127.57
converged:	True	LL-Null:	-267.55
Covariance Type:	nonrobust	LLR p-value:	4.616e-55

	coef	std err	z	P>|z|	[0.025	0.975]
const	-4.7289	1.298	-3.644	0.000	-7.273	-2.185
interceptions_thrown	-0.5463	0.220	-2.487	0.013	-0.977	-0.116
pass_fumbles_lost	-1.1304	0.426	-2.653	0.008	-1.965	-0.295
rush_attempts	0.1564	0.029	5.325	0.000	0.099	0.214
targets	-0.1040	0.034	-3.062	0.002	-0.171	-0.037
receiving_first_downs	0.2868	0.071	4.025	0.000	[illegible]	[illegible]
kickoff_yards	0.0084	0.002	4.322	0.000	[illegible]	[illegible]
kick_returns	-0.5976	0.112	-5.348	[illegible]	[illegible]	[illegible]
sack_yards	0.0497	0.017	2.890	[illegible]	[illegible]	[illegible]
interceptions	0.9724	0.284	3.419	[illegible]	[illegible]	[illegible]

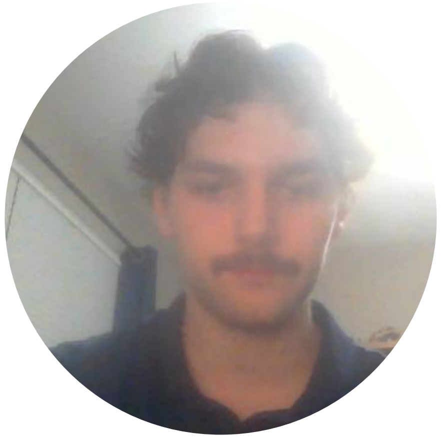
Testing Other Models

```
Logistic Regression Accuracy: 0.8846153846153846
False Positive Rate: 15.384615384615385
False Negative Rate: 7.6923076923076925

Random Forest Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255

XGBoost Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

Naive Bayes Accuracy: 0.8846153846153846
False Positive Rate: 17.94871794871795
False Negative Rate: 5.128205128205128
```

```
Logistic Regression Accuracy: 0.8461538461538461
False Positive Rate: 17.94871794871795
False Negative Rate: 12.82051282051282

Random Forest Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

XGBoost Accuracy: 0.8076923076923077
False Positive Rate: 17.94871794871795
False Negative Rate: 20.51282051282051

Naive Bayes Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255
```

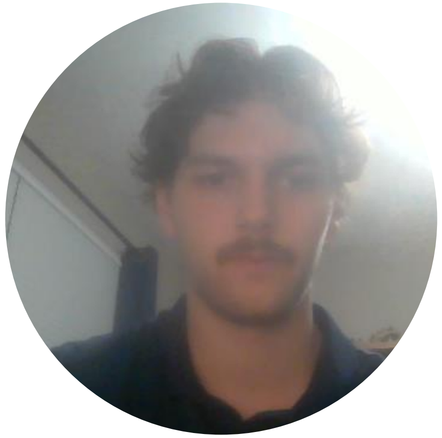
Conclusion

- Utilized exploratory data analysis to familiarize and identify certain features that could be an indicator to predict if an NFL team will win or lose.
- Modeled these features and identified significant features using logistic regression.
- Build four models on two different feature sets that resulted in strong accuracy score with decently low false positive and negative rates.



Further Study

- I would collect more data.
  - I wanted to use current data.
  - The downside is having a small dataset.
- I would also test out more models.
- Feature engineering with correlated features.



Data Source

- https://sports.yahoo.com/nfl/stats/weekly/?selectedTable=0&week={%22week%22:5,%22seasonPhase%22:%22REGULAR_SEASON%22}